

DATABASE MANAGEMENT SYSTEMS

Comprehensive Note: Weak Entity Set & Relational Mapping

ER Modeling

Relational Schema

Exam Study Guide

1. What is a Weak Entity?

A **Weak Entity** is an entity type in a Database Management System (DBMS) that **does not possess its own primary key** and cannot be uniquely identified using its own attributes alone.

It relies on another entity type, known as the **owner entity** or **strong entity**, for its complete identification.

Core Definition

Weak Entity = An entity that cannot be uniquely identified on its own and is identification-dependent on a Strong / Owner Entity.

Conceptual Example:

- **Employee:** Has a unique attribute *Employee_ID*, which uniquely identifies each record. (Strong Entity)
- **Dependent:** Contains attributes like *Name* and *Age*. Since multiple dependents across different employees can share the exact same name and age, *Dependent* cannot be identified independently. (Weak Entity)

To uniquely identify a dependent, we combine the owner's key with the partial key:

Identification Key = Employee_ID + Dependent_Name

2. Strong Entity vs. Weak Entity

Feature	Strong Entity	Weak Entity
Primary Key	Has its own primary key.	Does not have its own primary key.
Identification	Can be identified independently.	Cannot be identified independently.
Dependency	Exist independently without reliance.	Depends on an owner/strong entity.
ER Diagram Notation	Represented by a Single Rectangle .	Represented by a Double Rectangle .
Participation	Can have partial or total participation.	Always total participation in identifying relationship.

3. Owner Entity & Identification Dependency

The entity upon which a weak entity depends is called the **Owner Entity** or **Strong Entity**.

Employee (Owner Entity)

—— [Has] ——

Dependent (Weak Entity)

Identification / Existence Dependency: A weak entity exists in the database solely in relation to its owner entity. If an owner entity instance (e.g., *Employee E1*) is removed, all associated weak entity records (e.g., dependents of *E1*) must typically be removed as well because they lose their identification context.

5. Total Participation of Weak Entity

A vital rule in database modeling: **A weak entity ALWAYS has total participation in its identifying relationship.**

★ Important Rule for Exams

Every weak entity instance *must* be associated with an owner entity. Therefore, the link from the weak entity to the identifying relationship is always represented as a **double line (total participation)**. Conversely, the owner entity may have partial participation (an employee doesn't need to have dependents).

6. Partial Key (Discriminator)

Although a weak entity lacks a full primary key, it usually contains a **Partial Key** (also called a *discriminator*). A partial key is a set of attributes that can distinguish among weak entity instances that belong to the *same* owner entity.

Sample Data Illustration:

Employee_ID (Owner PK)	Dependent_Name (Partial Key)	Age
E1	<u>A</u>	16
E1	<u>B</u>	17
E2	<u>A</u>	16

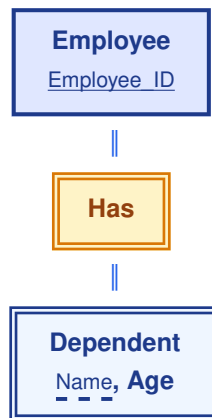
Analysis: *Dependent_Name* = "A" appears for both *E1* and *E2*. Thus, *Dependent_Name* alone is not globally unique. However, (*E1*, *A*) and (*E2*, *A*) are unique combinations across the entire table.

7. ER Diagram Notations Summary

Concept	ER Diagram Notation	Visual Representation / Description
Strong Entity	Single Rectangle	[Entity]
Weak Entity	Double Rectangle	[[Entity]]
Normal Relationship	Single Diamond	< Relationship >
Identifying Relationship	Double Diamond	<< Relationship >>
Key Attribute	Underlined Text inside Oval	<u>Employee_ID</u>
Partial Key (Discriminator)	Dashed Underline Text inside Oval	<u>Dependent_Name</u>
Total Participation	Double Line	=====

8. Identifying Relationship Notation

The relationship connecting a weak entity to its owner entity is called an **Identifying Relationship** (represented by a Double Diamond).



9. Conversion to Relational Schema (Tables)

A classic examination question asks: *How many tables are created when converting a Strong-Weak entity relationship into the relational model?*

Answer: Exactly **2 Tables** are required (not 3). The identifying relationship is merged into the weak entity table.

Relational Schema Definition:

```
-- Table 1: Strong Entity (Employee)
Employee (
    Employee_ID PRIMARY KEY,
    Name,
    Address
);

-- Table 2: Weak Entity with Owner Key (Has_Dependent)
Has_Dependent (
    Employee_ID,
    Dependent_Name,
    Age,
    PRIMARY KEY (Employee_ID, Dependent_Name),
    FOREIGN KEY (Employee_ID) REFERENCES Employee(Employee_ID) ON DELETE CASCADE
);
```

Why no separate table for the relationship?

Because the weak entity cannot exist without the owner, the primary key of the owner entity (***Employee_ID***) is directly migrated into the weak entity relation as a **Foreign Key**. Combining this Foreign Key with the weak entity's Partial Key (***Dependent_Name***) forms the complete **Composite Primary Key**.

10. Quick Revision & Exam Essentials ★

1. **No Primary Key:** Weak entity has no primary key of its own.
2. **Dependency:** Cannot be identified independently; depends on an owner entity.
3. **Total Participation:** Always participates totally in the identifying relationship.
4. **Partial Key:** Represented with a *dashed/dotted underline*.
5. **Notations:** Weak Entity = *Double Rectangle*, Identifying Relationship = *Double Diamond*.
6. **Table Conversion:** Requires 2 tables. Owner's PK becomes FK in weak entity relation.
7. **Composite PK:** *Composite Primary Key = Foreign Key (Owner PK) + Partial Key.*

🔥 Easy Memory Trick

"Weak = Can't Stand Alone"

No Own PK → Depends on Owner → Total Participation → Double Rectangle → Double Diamond → FK + Partial Key = Composite PK

★ One-Line Exam Definition

"A **weak entity** is an entity that lacks a primary key of its own and is uniquely identified by combining its partial key with the primary key of its identifying owner entity."